POMDP Formulation Approximations

POMDP Computational Complexity

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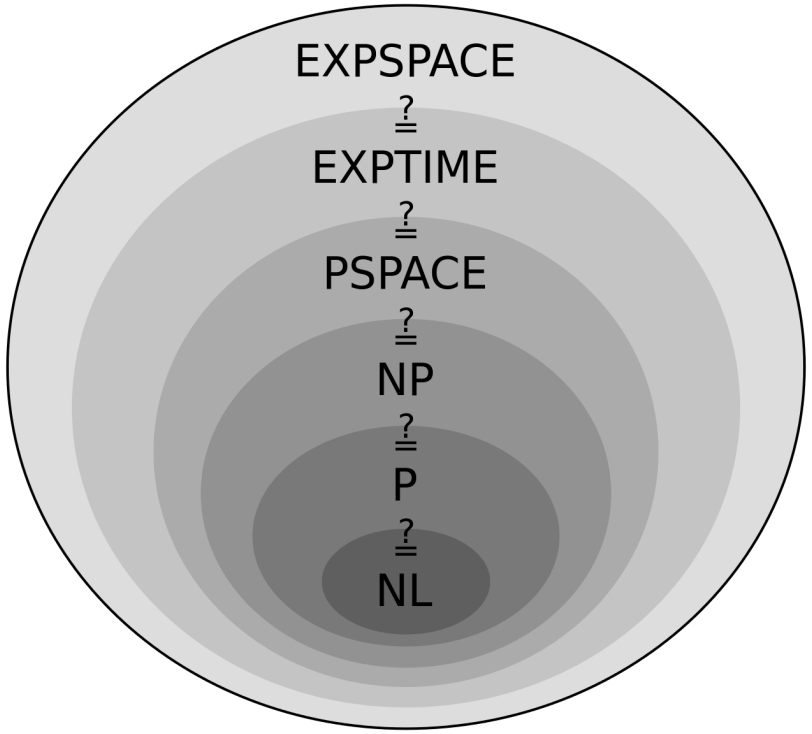

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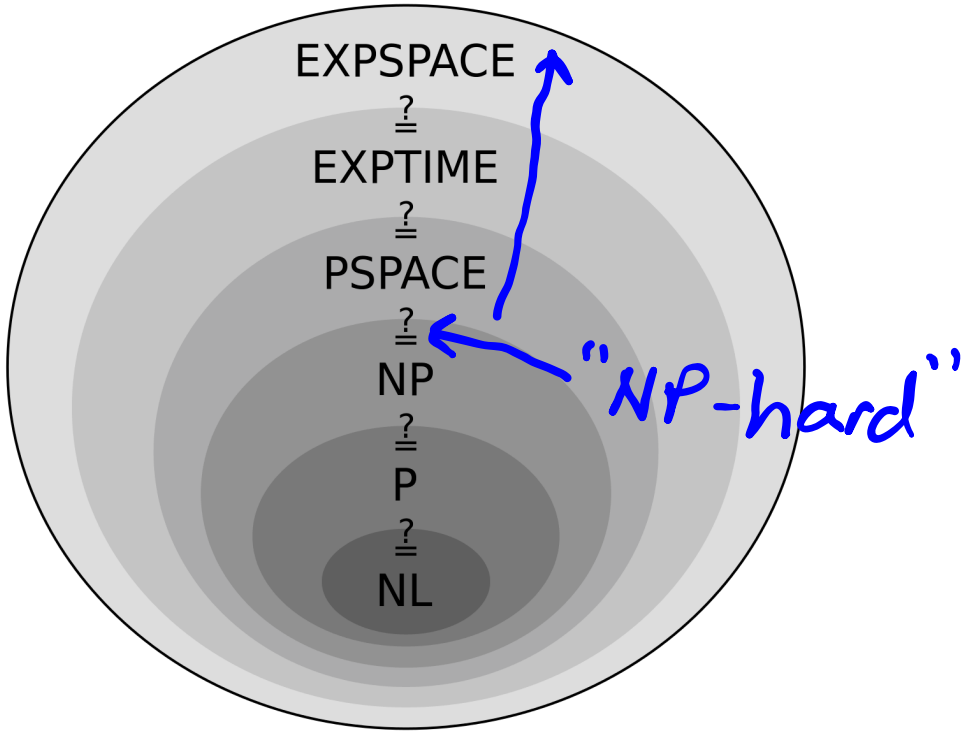

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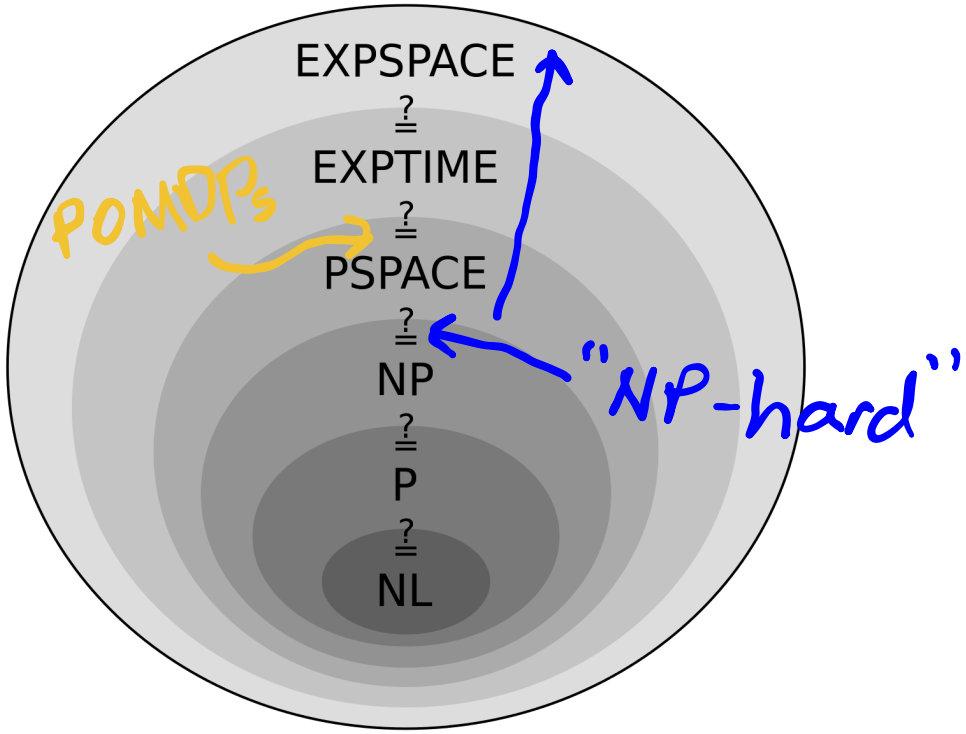

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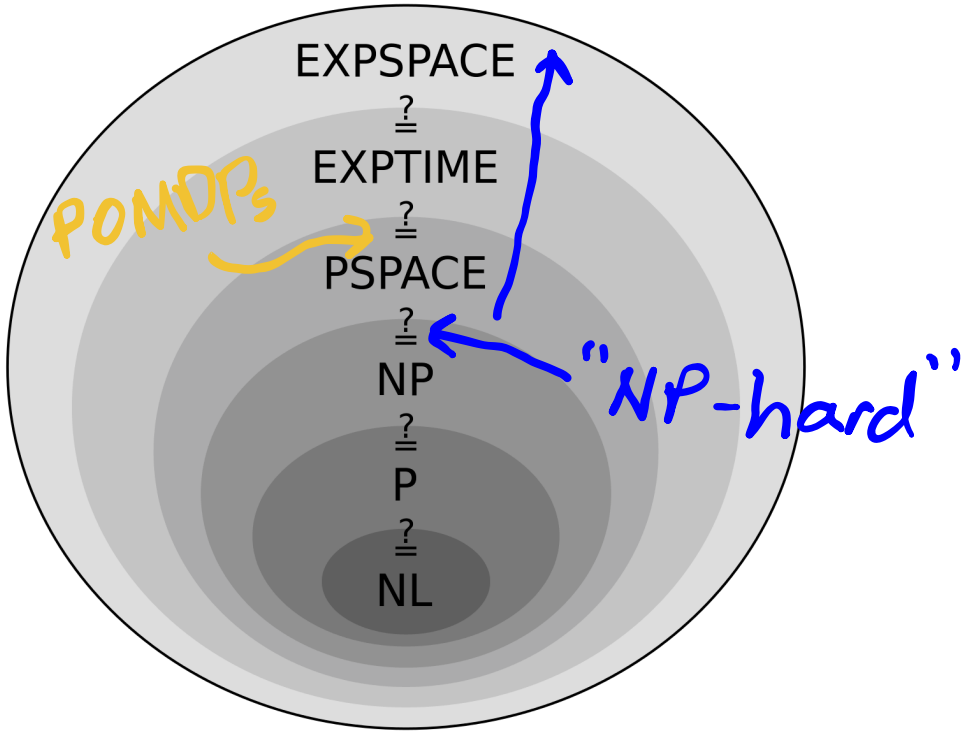

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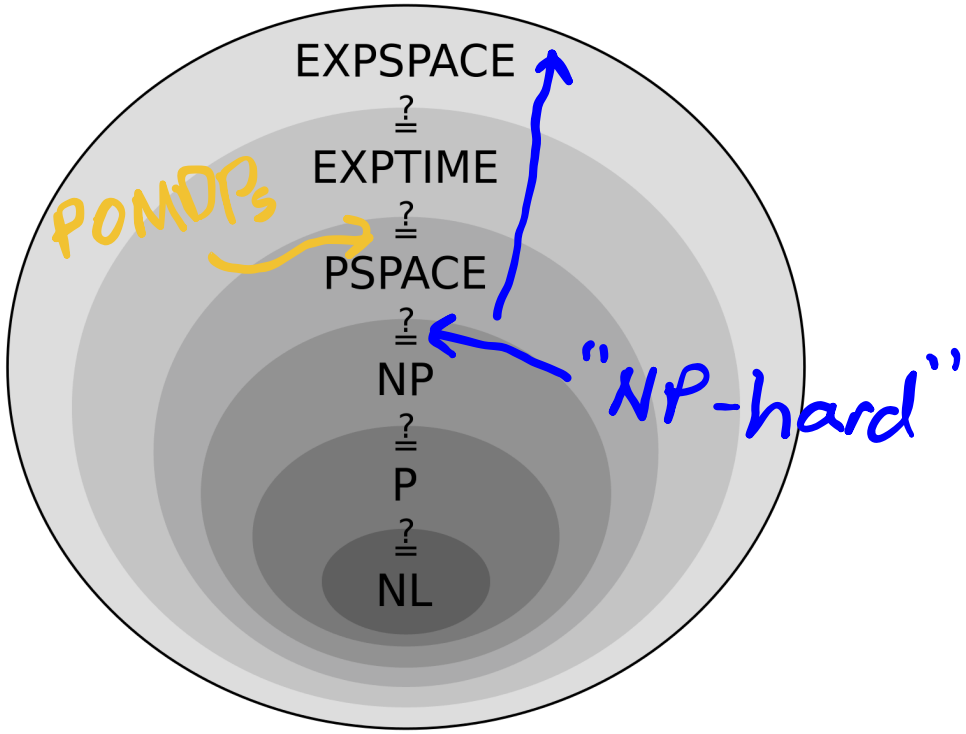

Approximate POMDP Solutions

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Numerical Approximations

(approximately solve original problem)

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Offline

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(solve a slightly different problem)

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Today!

POMDP Objective

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(Analogous to LQR MDP)

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<u>LQG POMDP</u>

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$$\pi^*_{\mathrm{LQG}}(b) = -K_{\mathrm{LQR}}\,\mu_b$$

QMDP

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$$\pi_{\mathrm{QMDP}}(b) = \underset{a\in A}{\operatorname{argmax}} \ \underset{s\sim b}{\mathrm{E}}\left[Q_{\mathrm{MDP}}(s, a)\right]$$

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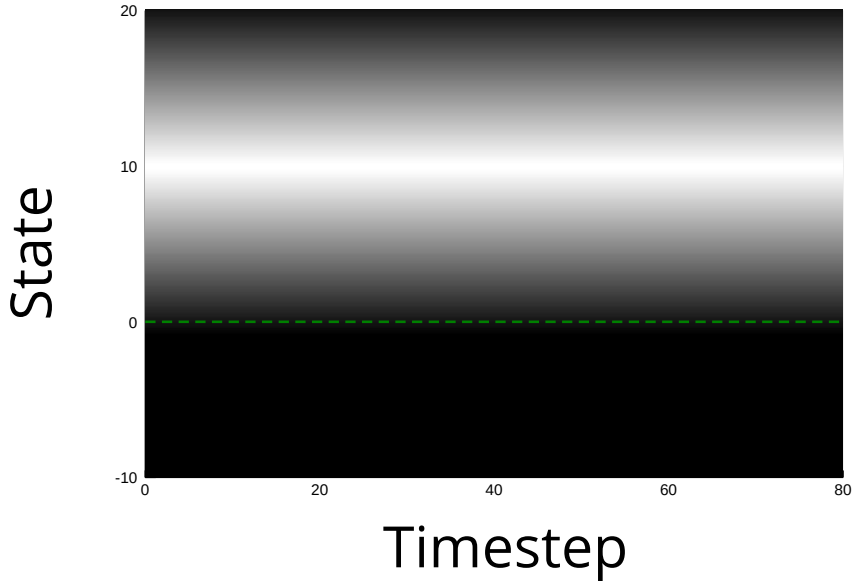
Example: Tiger POMDP with Waiting

POMDP Example: Light-Dark



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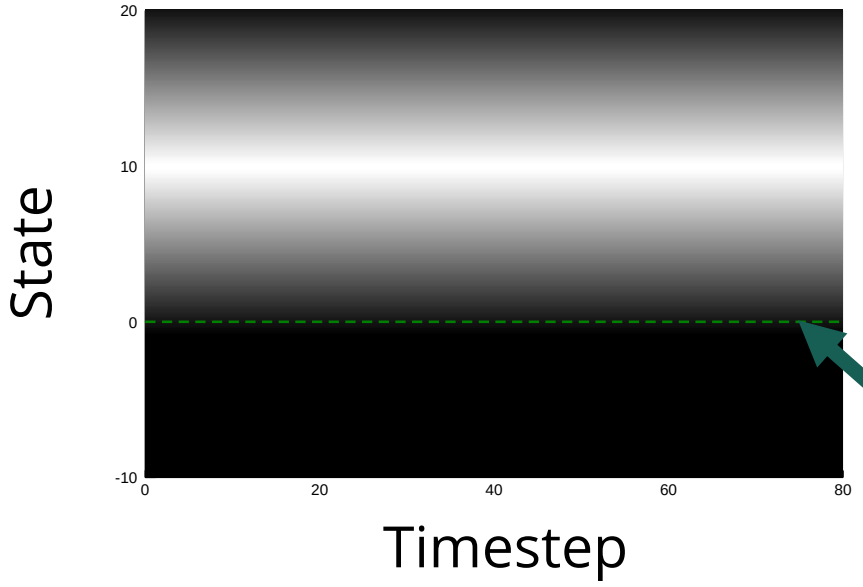

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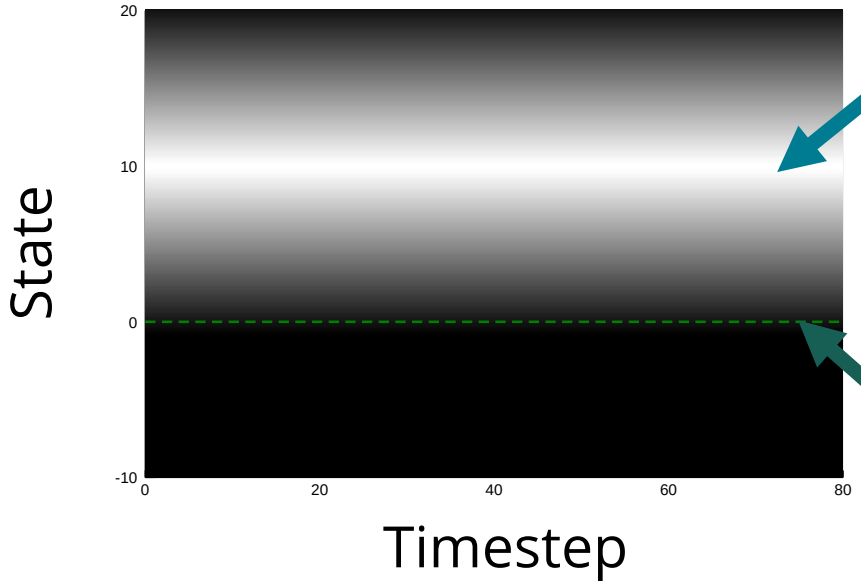

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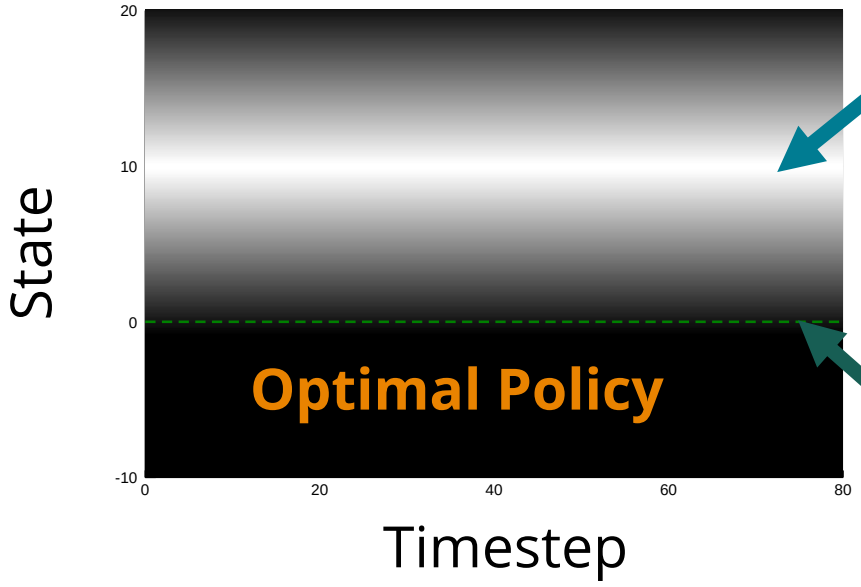

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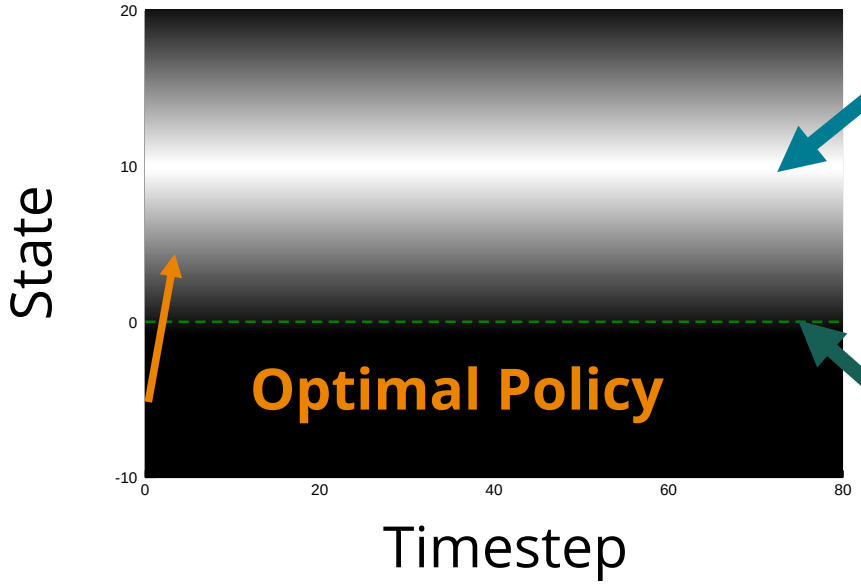

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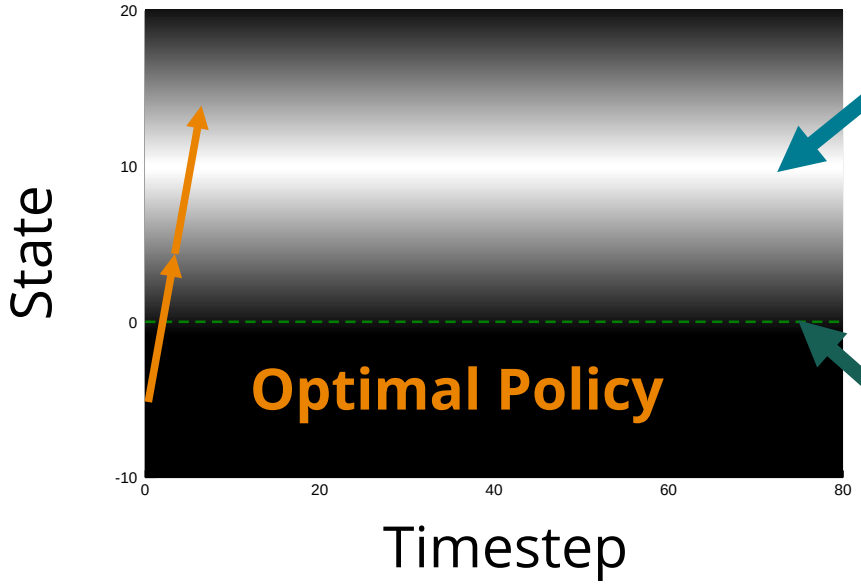

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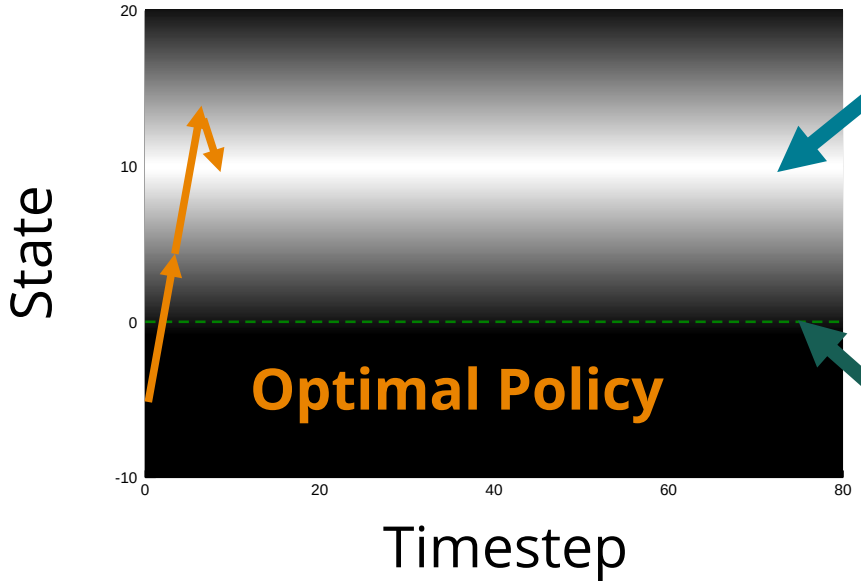

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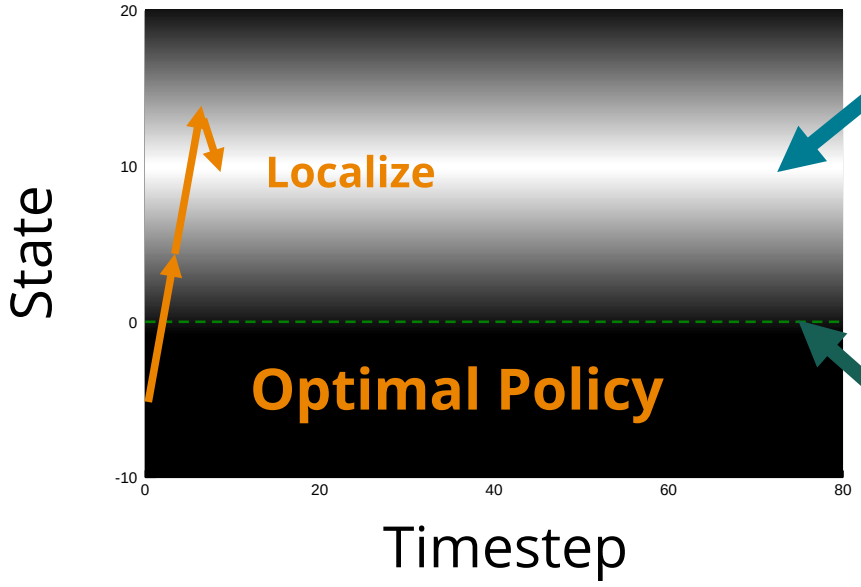

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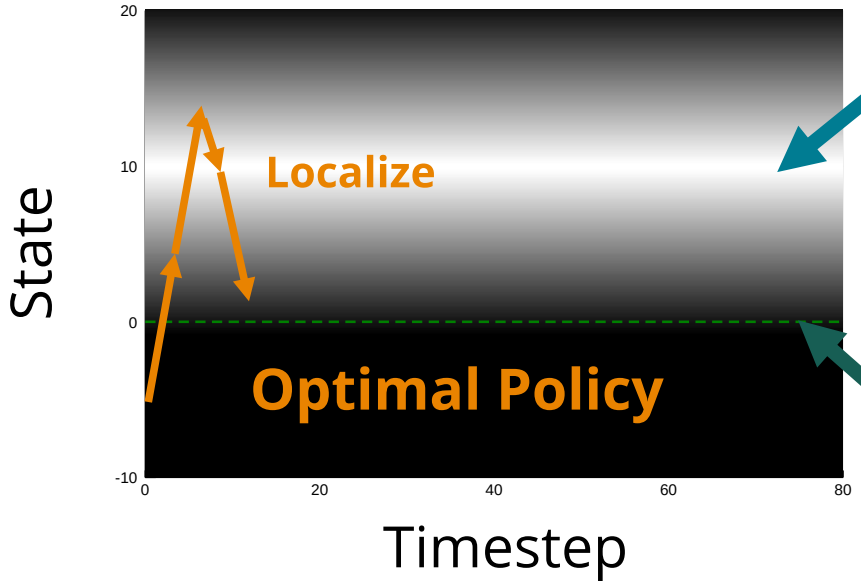

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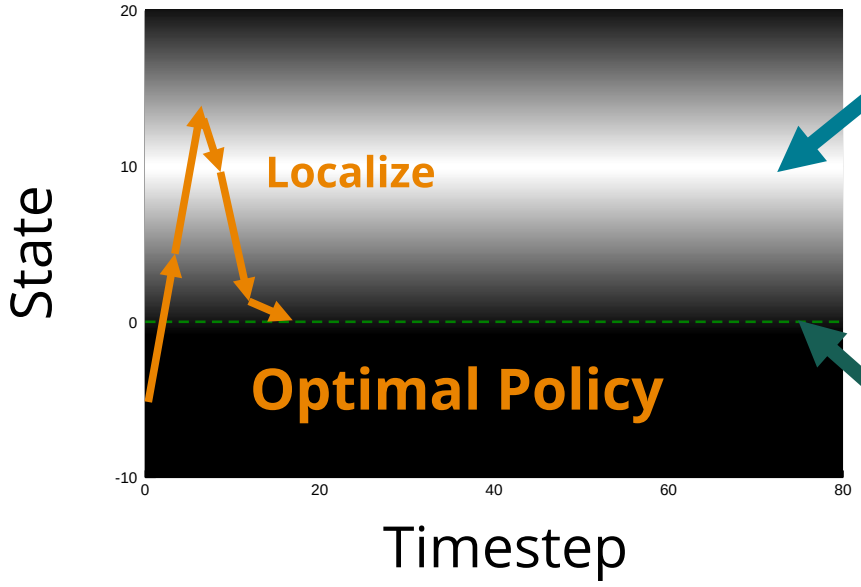

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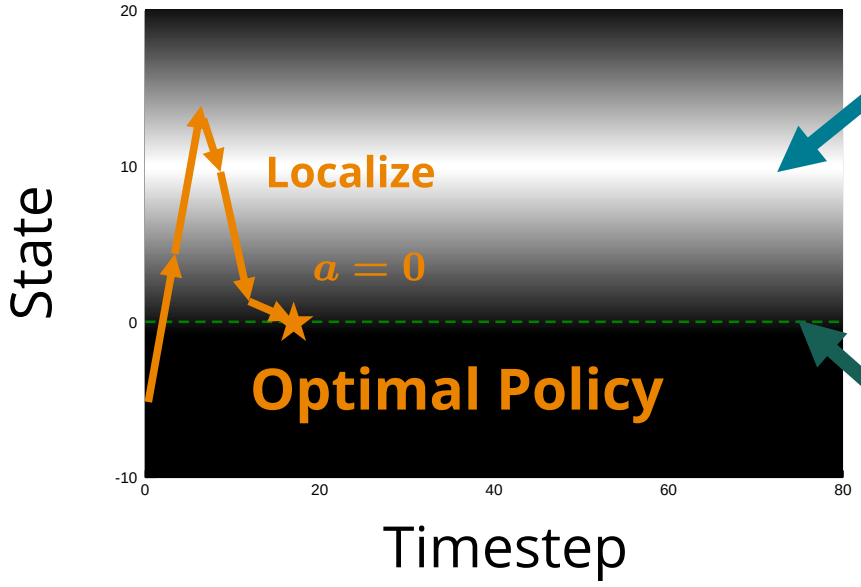

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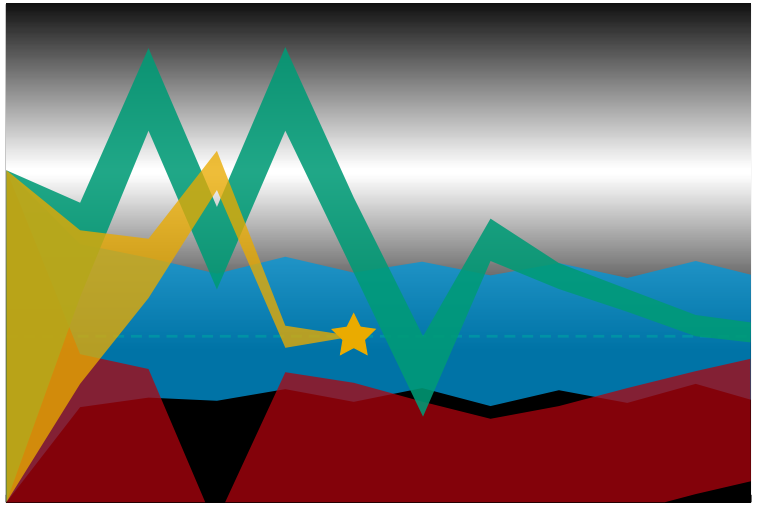

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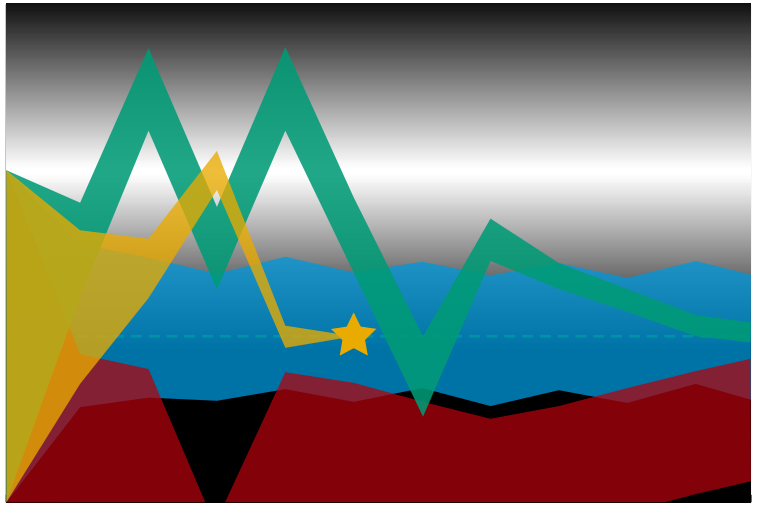

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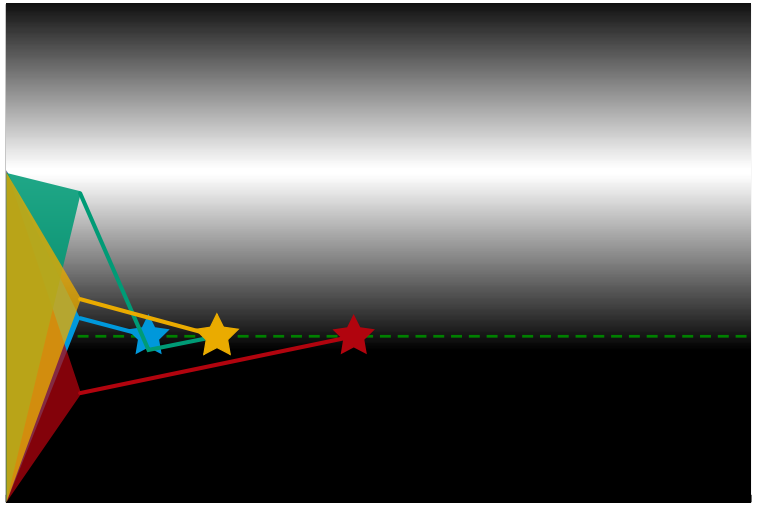

Same as **full observability** on the next step

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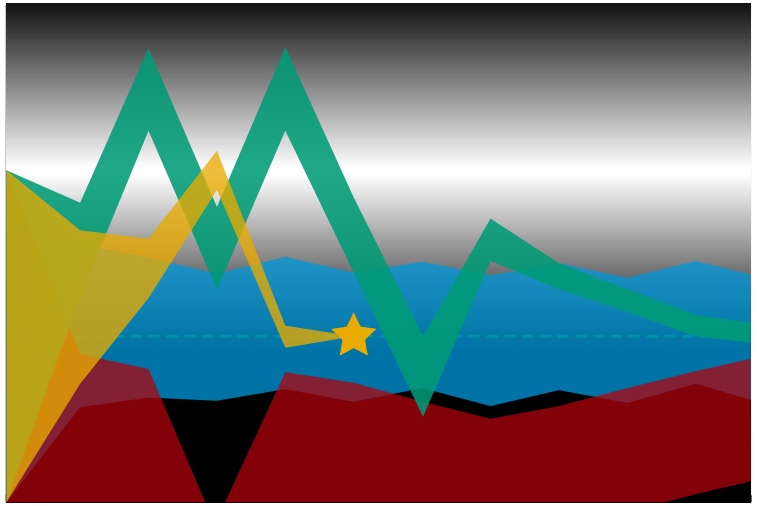

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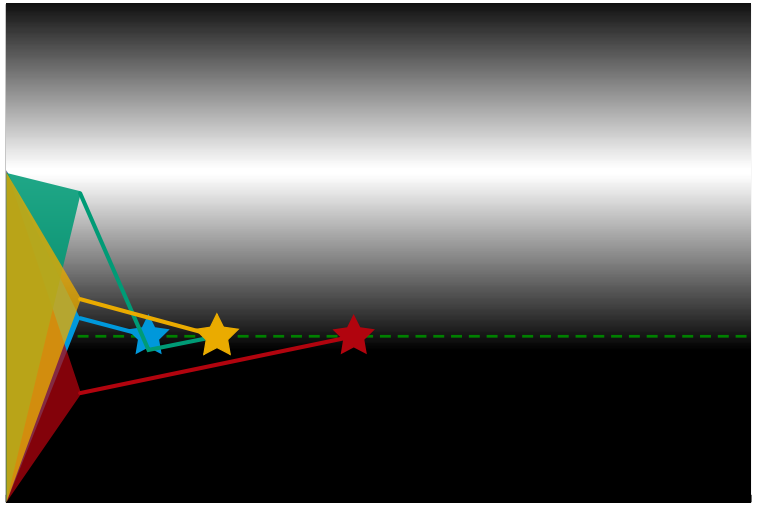
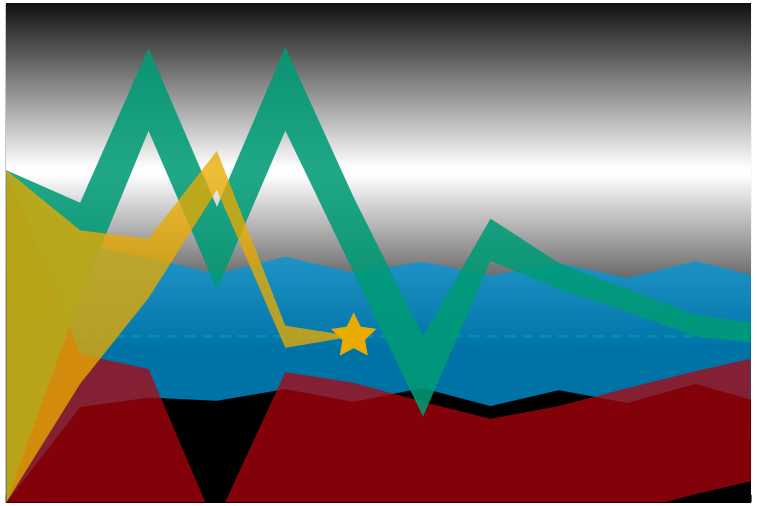


Same as **full observability** on the next step

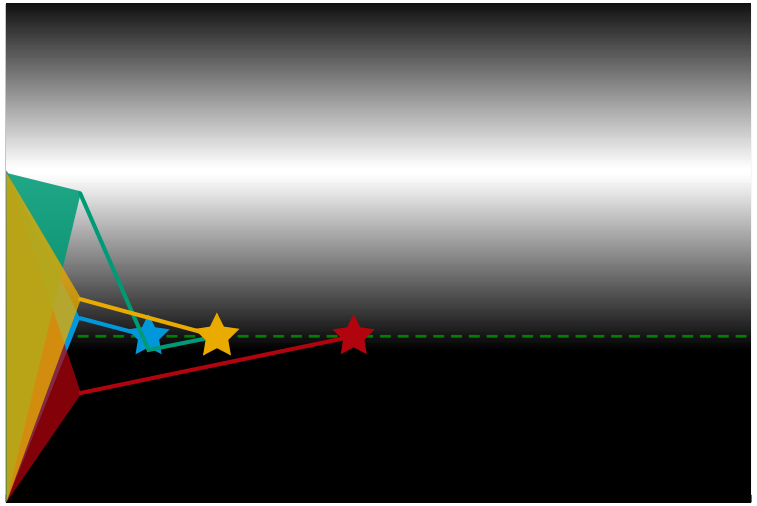
QMDP Behavior

(Break)

POMDP "Imagination"

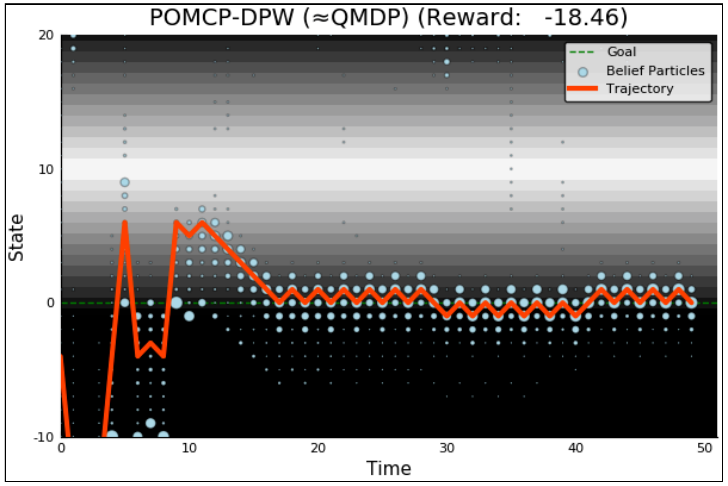


QMDP "Imagination"



QMDP Behavior

(Break)



Same as **full observability** on the next step

QMDP

Very effective for most POMDPs and cheap (same cost as solving MDP)

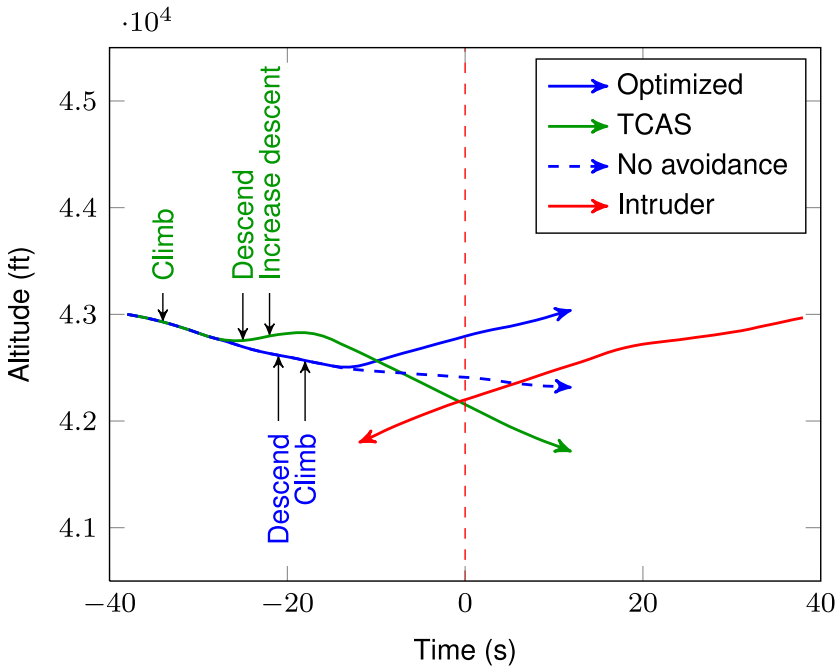
Two cases where it may not work well:

QMDP

Very effective for most POMDPs and cheap (same cost as solving MDP)

Used for ACAS X

[Kochenderfer, 2011]



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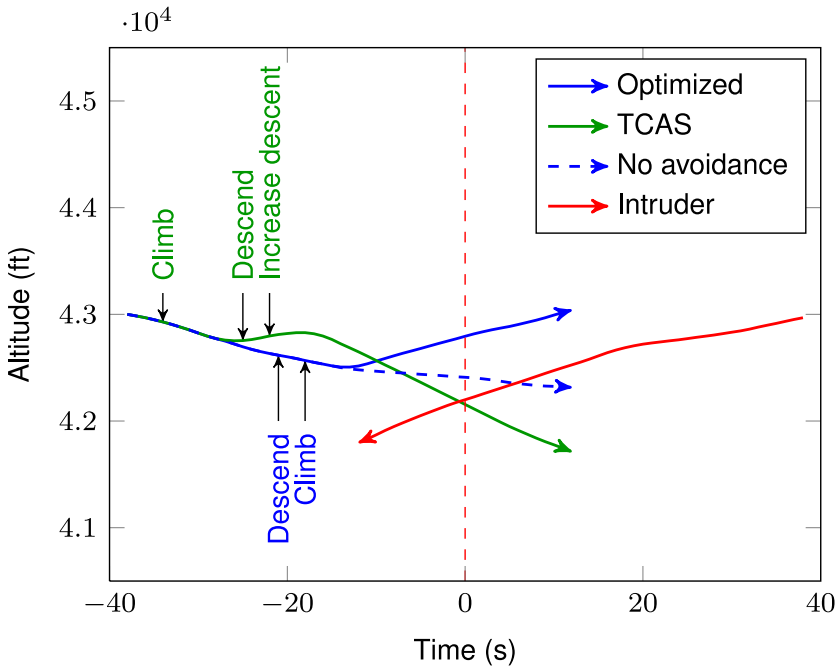

Two cases where it may not work well:

- *Costly* information gathering

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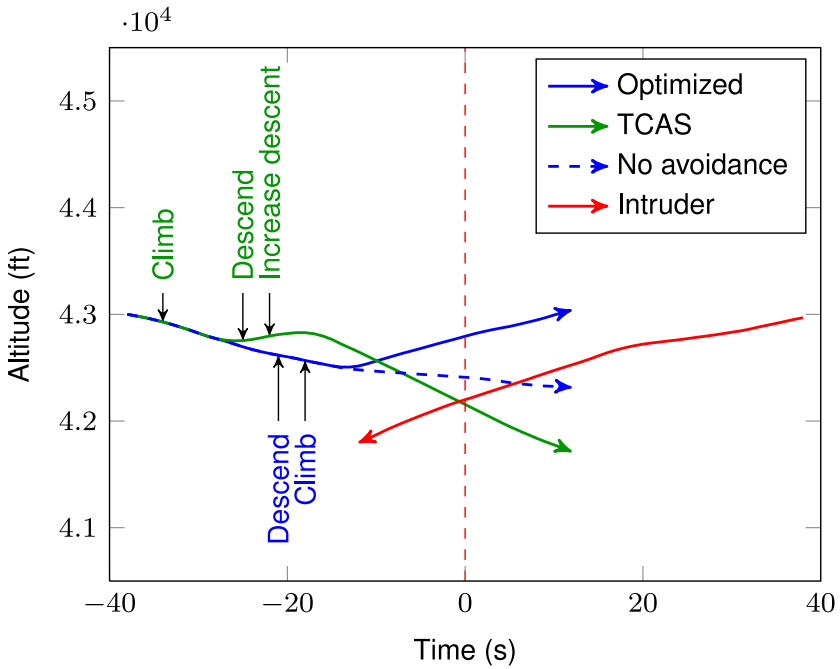

Two cases where it may not work well:

- *Costly* information gathering
- Long-lived uncertainty

Hindsight Optimization

POMDP Objective

$$\pi^* = \underset{\pi:B\to A}{\operatorname{argmax}} \ \mathrm{E}\left[\sum_{t=0}^{\infty} \gamma^t R(s_t, \pi(b_t))\right]$$

$$b' = \tau(b, a, o)$$

k-Markov

POMDP Objective

$$\pi^* = \underset{\pi:B\to A}{\operatorname{argmax}} \ \mathrm{E}\left[\sum_{t=0}^{\infty} \gamma^t R(s_t, \pi(b_t))\right]$$

$$b' = \tau(b, a, o)$$

Open Loop

A.K.A. Open Loop Feedback Control (OLFC)

POMDP Objective

$$\pi^* = \underset{\pi:B\to A}{\operatorname{argmax}} \ \mathrm{E}\left[\sum_{t=0}^{\infty} \gamma^t R(s_t, \pi(b_t))\right]$$

$$b' = \tau(b, a, o)$$